Paper2Agent

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Reimagining research papers as interactive and reliable AI agents

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Abstract

Here we introduce Paper2Agent, an automated framework that converts research papers into artificial intelligence (AI) agents. Paper2Agent transforms research output from passive artefacts into active systems that accelerate use and discovery. Conventional research papers require readers to understand and adapt the paper's code, data and methods to their work, creating barriers to dissemination and reuse. Paper2Agent addresses this challenge by converting a paper into an AI agent that functions as a virtual corresponding author, exposing its manuscript, supplementary materials, datasets, code and workflows as active, agent-native knowledge rather than static text. It analyses the paper and codebase using multiple agents to construct a model

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The Motivation

“papers are fundamentally passive objects: a reader must discover the paper (not an easy task given the flood of publications), parse its contributions and manually determine how to apply them to their own work. In particular, when a paper describes a new computational method, substantial technical barriers often remain before the method can be used on new data.”

“research outputs are passively siloed behind technical barriers. Paper2Agent reimagines research dissemination by turning static papers into active AI agents. Each agent serves as an interactive expert on the corresponding paper, capable of demonstrating, applying and adapting its methods to new projects.”

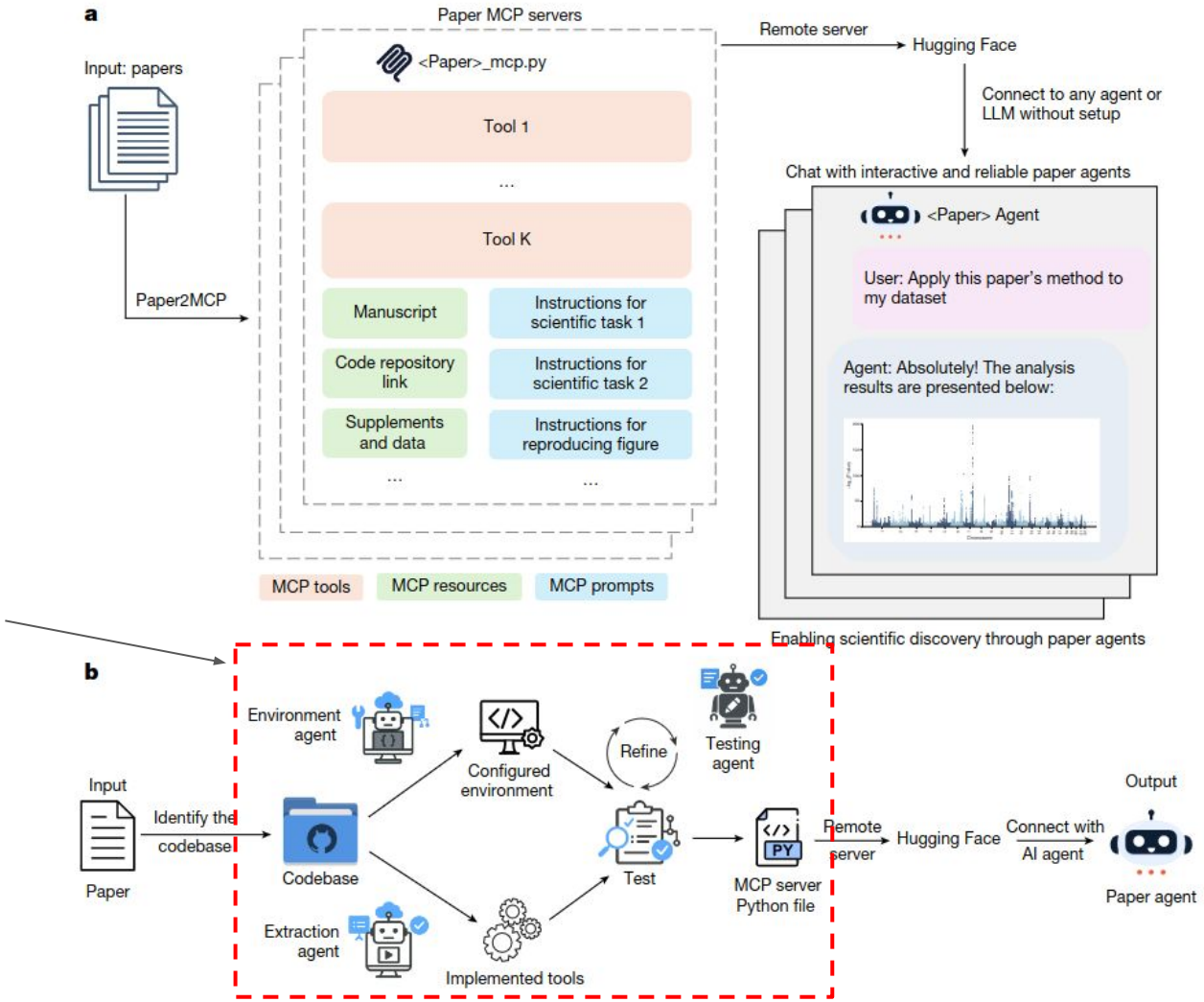
Model Context Protocols (MCP) and Skills

Dimension	Skills	MCP
Primary purpose	Teach an agent **how to perform a task**	Give an agent **capabilities/tools it can use**
Core question	*“How should I do this?”*	*“What can I do?”*
Contains	Instructions, methodology, workflows, rules, domain knowledge	Tools, APIs, data sources, resources
Example	“How to assess whether a paper is reproducible”	“Read a GitHub repository”
Agent interaction	Agent selects/loads a relevant skill	Agent selects and calls a tool
Typical content	SKILL.md, prompts, procedures, reference material	Tool definitions, schemas, API interfaces

Paper2MCP (which is a skill) turns your code into an MCP server so an agent can use it.

Then there idea is you make this MCP public, and anyone can access it and use your paper interactively.

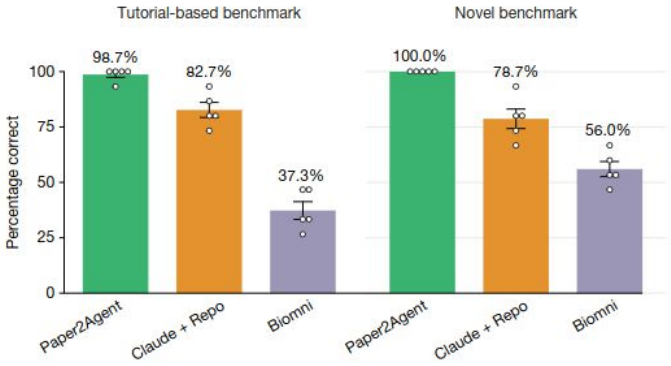
Can ask CC to do stuff like: Implement the method in this paper but with a dataset I have here

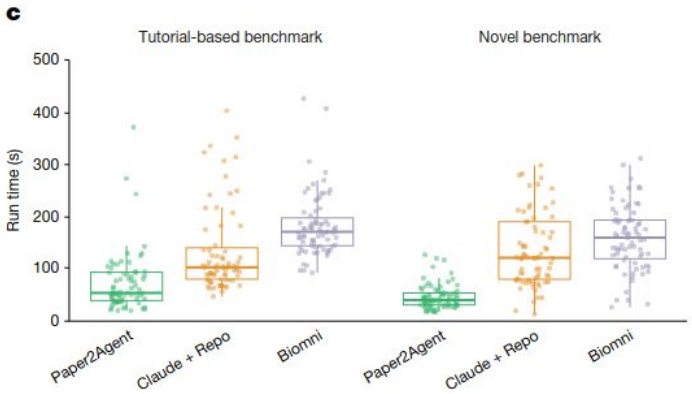


How did they validate it?

Dived into 2 example in detail: AlphaGeonome and Scanpy

For AlphaGenome: “we benchmarked the AlphaGenome agent against human- executed ground truth, Claude Code with direct repository access (Claude + Repo) and Biomni (https://biomni.stanford.edu/) (Fig. 2b).”





The Paper2Agent-generated agent did not have access to the manuscript or raw repository during evaluation,

https://github.com/jmiao24/Paper2Agent/tree/main

Lets look at their repo.

I tried it out on a toy example . .

“Use the paper2agent skills to convert https://github.com/alan-turing-institute/autoemulate into MCP tools in this project directory.
For now, only focus on the functionality needed to run the tutorial at https://github.com/alan-turing-institute/autoemulate/blob/main/docs/tutorials/emulation/01_quickstart.ipynb”

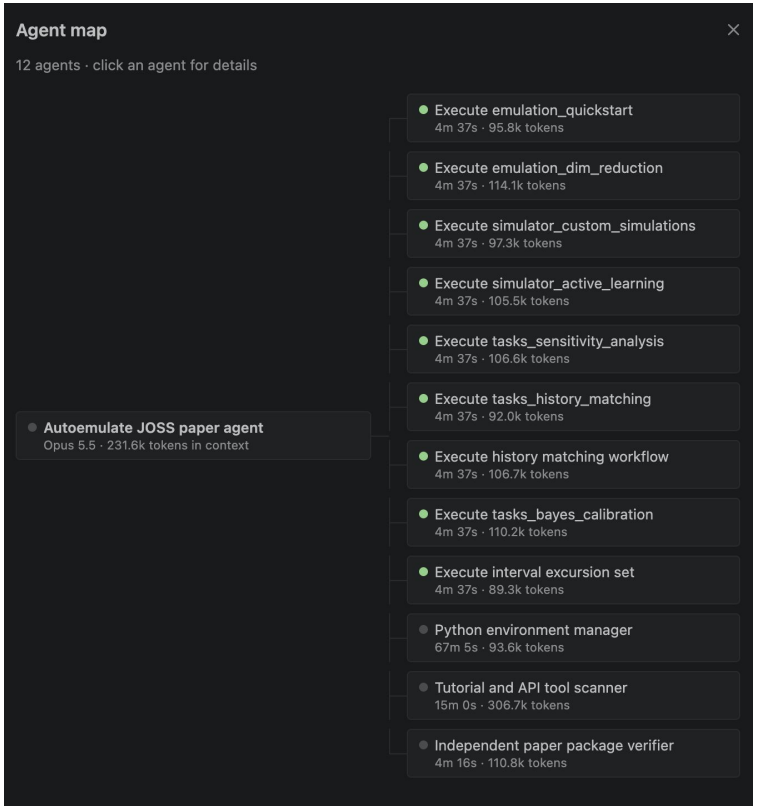
- Builds a 5 tool MCP server for my quickstart

- Then becomes very quick to fiddle with the tutorial, changes parameters, extending it, etc.

Large Scale Evaluation

Among the 100 computational biology papers, 74 were successfully agentified, yielding 599 proposed tools, of which 593 passed automated validation. The major failure modes among the remaining papers were missing executable code, missing data or model artefacts, environment or dependency failures, and non-generalizable scripts. On 300 tutorial-derived benchmark questions, Paper2Agent with Sonnet 4 achieved 91.2 ± 1.6% accuracy, outperforming Claude Code with direct repository access using Sonnet 4 (80.3 ± 2.3%; $P < 0.0001$) and Sonnet 4.6 (86.3 ± 1.1%; $P < 0.0001$). Paper2Agent also reduced query cost and latency (US $0.20 and 1.6 min per query, compared with US $0.38 and 4.3 min for using Sonnet 4 directly with the paper and repo). Across 42 execution-based tasks from 10 non-biology computational papers, Paper2Agent achieved 98.1 ± 0.8% accuracy across 5 independent runs, demonstrating generalization beyond computational biology. Paper2Agent also remained useful when executable tools could not be constructed. Across 26 data- and discovery-focused papers, the resource layer achieved 89.0 ± 3.1% accuracy on 100 synthesis-based questions, outperforming a Claude browser-use baseline (82.0 ± 3.8%; $P = 0.03$) while being 34× cheaper and 15× faster.

I tried it out a bigger example.

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“Agentify this paper: https://www.theoj.org/joss-papers/joss.10087/10.21105.joss.10087.pdf . The codebase is in this dir: . . .

Entire Auto Emulate codebase - ~200K LOC

Takes forever to build the MCP server

What I learned is its better suited to research papers with code provided rather than developed codebases.

1st attempt hit my token limit

2nd attempt, hit 3 hours and gave up. .

Discussion

Do we want this?

Is this an efficient way of dissemination?

I understand why this is quicker, but why can’t Claude Code approach get to the same level of accuracy?

Authors suggest using this approach as a ‘check’ for whether a paper is ‘ready for reproducibility’.

Paper2MCP

MCP tools are executable functions that encapsulate a paper's methodological contributions.

MCP resources serve as a repository of static assets, including the manuscript text, the associated codebase and supplementary materials such as datasets, tables and figures.

MCP prompts contain concise instructions that guide AI agents through complex, multi-step scientific workflows derived from a paper's text or codebase.

